

Web and Multimedia Technologies

Lab class 6 – JavaScript

Instructions to use the laboratory computers (room B2/B4):

1. If the computer is off, start it with Windows (all computers have a Linux-Windows dual boot)
2. Your username is *cl* followed by the six digits of your student ID; your password is *inf* followed by the last two digits of your student ID (e.g., if the student ID is 123456, the username is *cl123456* and the password is *inf56*)
3. At the end of the class, please log out

Steps:

1. Create the folder `javascript` and, within it, the file `index.html` containing the following HTML code (the value of the field *so, you are or are going to be* is calculated). Important: JavaScript is case-sensitive

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8">
    <title>Registration form</title>
    <style>
      #formstruct {
        display: table;
      }
      #labels, #inputs {
        display: table-cell;
      }
      input { padding: 0px; margin: 0px; }
    </style>
  </head>
  <body>
    <h1 style="font-family: sans-serif">Registration Form</h1>
    <form name="registration" id="registration" action="process.php">
      <div id="formstruct">
        <div id="labels">
          <label name="surname">Family name&nbsp;&nbsp;&nbsp;</label><br>
          <label for="name">First name</label><br>
          <label for="birthyear">Birth year</label><br>
        </div>
        <div id="inputs">
          <input name="surname" id="surname" size="20"><br>
          <input name="name" id="name" size="20"><br>
          <input name="birthyear" id="birthyear" size="3"><br>
        </div>
      </div>
      (so you are or are going to be&nbsp;&nbsp;&nbsp;
      <input name="years" id="years" size="1" readonly><br><br>
      <input type="button" value="Send">
    </form>
  </body>
</html>
```

2. In the `<head>` section, within the `<script>` tag, add the following JavaScript functions, so that, when you enter a birth year into the *Birth year* field and “click elsewhere” (onchange event handler), the page behavior is as follows:
 - a. If the input string is not an integer number, an error message box is displayed, and the *Birth year* field is cleared.
 - b. If the input string is an integer number, the resulting age appears in the field *so you are or are going to be* (= current year - content of the field *Birth year*).

```
function integerNumber(input) {
    var answer = true;
    if ( ! parseInt(input) ) {
        answer = false;
    } else {
        for (var i=0; i<input.length; i++) {
            if ((input.charAt(i) != "0")
                && ( ! parseInt(input.charAt(i))) ) {
                answer = false;
                break;
            }
        }
    }
    return answer;
}
```

The attribute `onchange="calculateYears() "` must be added to the `<input>` defining the *Birth year* field

```
function calculateYears() {
    if (! integerNumber(document.registration.birthyear.value)) {
        alert("Wrong birth year!");
        document.registration.birthyear.value = "";
        return false;
    } else {
        var date = new Date();
        document.registration.years.value = date.getFullYear() -
                                            document.registration.birthyear.value;
        return true;
    }
}
```

(Note that the function returns `true` or `false` depending on whether the value of the *Birth year* field is correct or not)

3. Still within the `<script>` tag in the `<head>` section, add the following function to check whether a field of the form is empty:

```
function exists(input) {
    var atLeastOneChar = false;
    if (input) {
        for (var i=0; i<input.length; i++) {
            if (input.charAt(i) != " ") {
                atLeastOneChar = true;
                break;
            }
        }
    }
    return atLeastOneChar;
}
```

4. Define a `checkForm()` function so that, when the *Send* button is pressed, the form is analyzed to check whether the fields *Family name* and *First name* are empty and the field *Birth year* actually contains an integer number (also calculating the age, like in Step 2). In case of errors, the function must display a proper message; otherwise, the form can be sent to the server through `document.registration.submit()`:

```
function checkForm() {
    var mes = "";
    if (! exists(document.registration.surname.value))
        mes = mes + "Missing family name\n";
    if (! exists(document.registration.name.value))
        mes = mes + "Missing first name";
    if (mes != "")
        alert(mes);
    res = calculateYears();
    if ((mes == "") && (res))
        alert("Ok, correct!\nThe form should now be sent to " +
            "the server through " +
            "document.registration.submit()");
}
```

The attribute `onclick="checkForm() "` must be added to the `<input>` defining the *Send* button

5. At the beginning of the page, add the current date and time by means of the following code (just after the `<body>`):

```
<script>
    var today = new Date();
    document.write("<h2>Date: " + today.getDate() + "/" +
        (today.getMonth() + 1) + "/" +
        today.getFullYear() + ", Time: " +
        today.getHours() + ":" +
        today.getMinutes() + "</h2>");
</script>
<hr width="40%" align="left">
```

6. At the end of the page, add the following HTML5 code:

```
<canvas width="200" height="200" id="canv"></canvas>
```

In the `<script>` tag within the `<head>` section, write the following function that draws a square and a rectangle (and call the function in the `onclick` event handler of the `<input>` defining the *Send* button, besides `checkForm()`):

```
function draw() {
    canvas = document.getElementById('canv');
    ctx = canvas.getContext('2d');
    ctx.fillStyle = '#0101fe';
    ctx.fillRect(50, 60, 100, 100);
    ctx.fillStyle = "rgba(220, 223, 0, 0.5)";
    ctx.fillRect(90, 105, 100, 80);
}
```

Now, the two graphical elements will be drawn when the *Send* button is pressed.

7. Move all the functions defined in the previous steps to the external file `control.js`, and load this file from `index.html` (through the `src` attribute of the `<script>` tag)